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To the Editor-in-Chief and Associate Editor
IEEE Transactions on Information Theory

Dear Editor:

Please consider the enclosed manuscript, “Rate, Leakage, and Distortion for Gaussian Sources: Water-Filling on a Tilted Weight,” for publication in the IEEE Transactions on Information Theory. I am the sole author. The review manuscript is in single-column format.

An encoder observes a jointly Gaussian vector and describes part of it to a decoder under a weighted quadratic distortion, while a second party holds a noisy linear observation of the source and never decodes. The description then has two costs, its rate and its conditional leakage to the second party. The single-letter region is known from secure source coding (Villard and Piantanida [1]), and Ekrem and Ulukus [2] and Xu, Chen, Chen, and Jin [3] treat vector Gaussian sources under a matrix distortion on the whole source, a setting in which their bounds already imply that one description minimizes both costs (Proposition 4). The manuscript determines the structure of the region under a weighted distortion on part of the source, where the two costs trade off.

The main results are in three groups. First, both costs are convex in the error covariance, and every frontier point is the fixed point of reverse water-filling on the decoder’s weight tilted by the second party’s information (Theorem 7), computable by a monotone iteration (Proposition 11). Second, explicit solutions. When the optimal description is one-dimensional, which an explicit certificate decides, the minimum leakage is a closed-form function of the largest eigenvalue of a matrix pencil in which the budget enters (Theorem 16). For a scalar second-party observation the frontier reduces to two scalar equations, and for a scalar described variable to one scalar root (Corollary 9 and Theorem 18). Third, directions. The best one-dimensional description turns with the budget unless the two parties’ geometries share it (Theorem 20), a description with a fixed direction pays an explicit excess leakage (Theorem 25), and the rate-optimal and leakage-optimal descriptions differ exactly under a stated misalignment condition (Theorem 22). With side information at the decoder, the manuscript proves the region for the continuous source (Theorem 29), settles the two degraded cases (Theorem 31), and brackets the general case between a Gaussian inner bound and two outer bounds that are tight in both degraded cases (Proposition 32). An appendix shows the same effects in fixed-rate scalar quantizers.

Several earlier documents are relevant, and they are distinct from one another. First, this Transactions declined a different manuscript of mine, IT-26-0996, on 27 July 2026 without review, on scope and presentation. That manuscript was a broad synthesis. The present submission is a focused source-coding paper, shares no theorem with IT-26-0996, and uses standard source-coding terminology throughout.

Second, the manuscript cites two archived manuscripts by the author, as disclosed in its Related Work section. Reference [21] (<https://doi.org/10.5281/zenodo.21776291>) is a preliminary manuscript containing the discrete region theorem, which coincides with [1, Theorem 3] and is credited there, and the scalar Gaussian corner. Reference [22] (<https://doi.org/10.5281/zenodo.22678341>) is a

manuscript on a scalar described variable and a scalar context, which was prepared for this Transactions but not submitted. Its closed form appears here as Corollary 17, credited to it, and its other Gaussian results are special cases of the present theorems. The present manuscript supersedes the Gaussian results of [22], which will not be submitted separately. Its binary rate–leakage frontier is not part of the present manuscript. A companion draft on vector described variables is likewise superseded and will not be submitted.

The present manuscript’s sources, verification scripts, and recorded outputs are archived at <https://doi.org/10.5281/zenodo.23127802> and at a pinned commit of the public repository named in its Appendix. The identities and instances are checked numerically and symbolically, and the scalar steps, the linear-algebra step of the rank bound, and the convexity of both costs in the error covariance are machine-checked in Lean 4, as described in the Appendix. The check scripts, their records, and the Lean modules are submitted as supplementary material. Also submitted, as optional supplementary reading, is a short note that relates the results to a broader programme on observer-dependent descriptions and reports a registered empirical campaign. Its measurements use a classifier cross-entropy proxy for leakage and are reported as evidence about the predictive usefulness of the Gaussian model, not as measurements of conditional entropy. The manuscript does not depend on it.

The use of AI tools in preparing the manuscript is disclosed in its Acknowledgment, as IEEE policy requires. This manuscript is not under review elsewhere, and I have no conflicts of interest to declare.

Sincerely,

Andrew H. Bond
Senior Member, IEEE